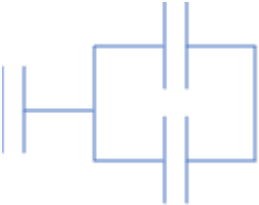


SECTION A

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		Capacitance equation used i.e. $C = \frac{\epsilon_0 A}{d}$ (1) Area = d^2 and cm conversion ($7.569 \times 10^{-3} \text{ m}^2$) (1) Correct separation ($2.91 \times 10^{-4} \text{ m}$ or 0.291 mm etc.) (1)	1	1 1		3	2	
	(b)		Using $Q = CV$ e.g. 87 [V] or substituting (1) Using the energy equation i.e. $\frac{Q^2}{2C}$ (1 st mark implied) (1) Final answer = $2.035 \times 10^{-8} \text{ [C]}$ or 20.35 n[C] hence correct (1) e.g. $0.87 \text{ }\mu\text{J} \approx 0.9 \text{ }\mu\text{J}$ Accept $0.87 \text{ }\mu\text{[J]}$, 222 p[F]			3	3	3	
	(c)		Same pd or full pd [to each C] (1) Same charge [on each C] OR $C = \frac{Q_{\text{total}}}{V} = \frac{Q + Q}{V}$ OR $C = \frac{Q_1 + Q_2}{V}$ (1) Hence double charge OR $C = \frac{Q}{V} + \frac{Q}{V}$ OR $C = \frac{Q_1}{V} + \frac{Q_2}{V}$ i.e. seeing that capacitances can be added (1)		3		3		
	(d)		$2 \times 230 \text{ pF}$ in parallel and then 230 pF in series 		1		1		
			Question 1 total	1	6	3	10	5	0